COURSE REVIEW 1

General Chemistry I

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Calculations Involving Sigfigs

- Uncertainty of the result should reflect the uncertainties of all input numbers

$$35.2 + 0.73 = (35.2 \pm 0.2) + (0.73 \pm 0.02)$$
$$= 35.93 \pm 0.22 = 35.9$$

+ and −: Round off to the leftmost decimal place

$$35.2 \times 0.73 = (35.2 \pm 0.2) \times (0.73 \pm 0.02)$$
$$= [24.85, 26.55] = 26$$

× and ÷: Round off to the smallest number of sigfigs

Chemical Formulas

- Empirical formula

Elemental composition → Empirical formula → Molecular formula

  - "CH" → C_2H_2, C_6H_6, etc.
  - "$AlCl_3$" → Al_2Cl_6, "P_2O_5" → P_4O_{10}

- Nomenclature

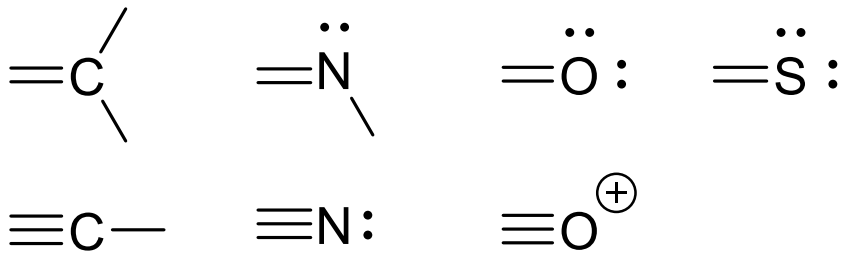
$FeCl_2$

- 二氯化铁 Iron dichloride
- 氯化铁(II) Iron(II) chloride
- 氯化亚铁 Ferrous chloride

mono- 单
di- 二
tri- 三

Number of Valence Electrons

Electrons	0	1	2	3	4	5	6	7	8
Valence	0	1	2	3	4	3	2	1	0
Examples	H^+			C^+	N^+	O^+			
		H		B	C	N	O	F	Ne
					B^-	C^-	N^-	O^-	F^-



Quantum Mechanics

- Wave–particle duality
  - Light can behave like particles
  - Photoelectric effect
  - Electrons can behave like waves

$$E_{\text{photon}} = h\nu = \frac{hc}{\lambda}$$

$$\lambda_{\text{electron}} = \frac{h}{p}$$

- Wave functions
  - Describes the motion of both light and electrons
  - Its square equals probability
  - Its number of nodes indicates total energy

n = Total number of nodes + 1

$l = 0, 1, 2, \ldots, n - 1$

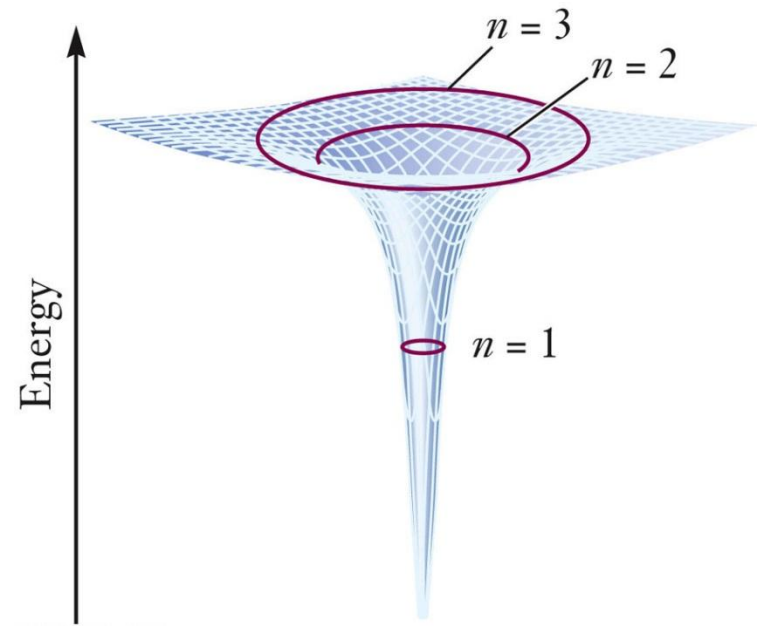
$m = -l, -l + 1, \ldots, l - 1, l.$

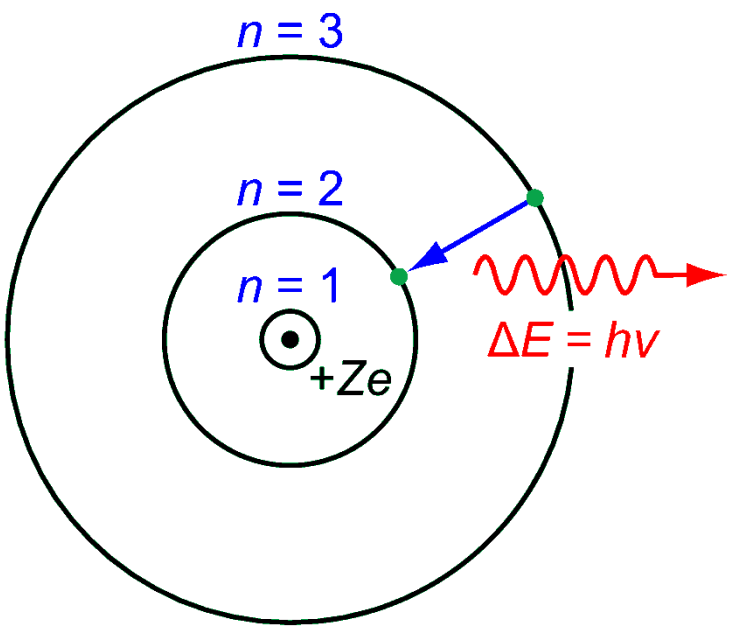
The Hydrogen Atom

- Bohr model

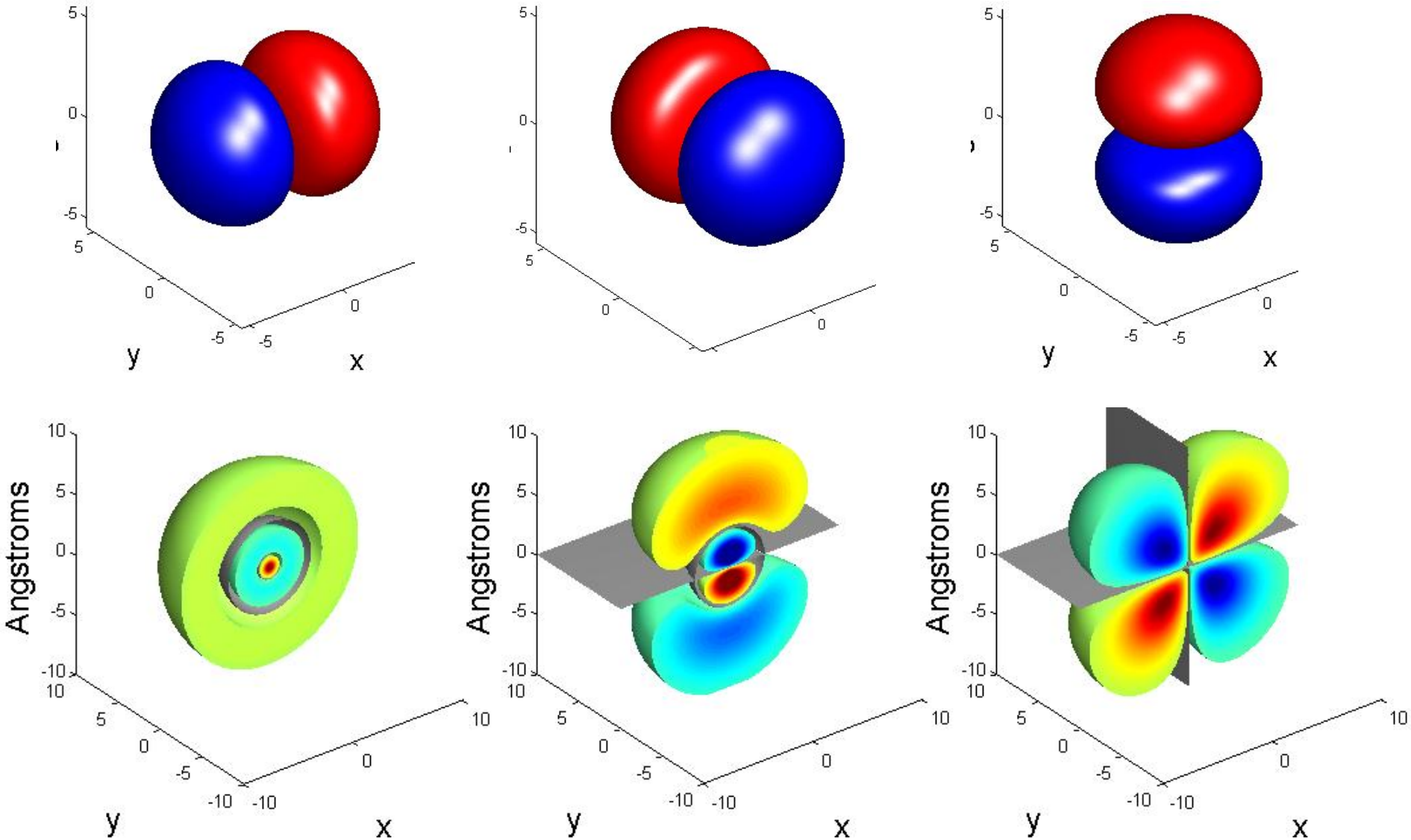
$$E = -\mathrm{Ry}\frac{Z^2}{n^2},\ \mathrm{Ry} = 13.6\ \mathrm{eV}$$

$$r_n = \frac{n^2}{Z}a_0,\quad a_0 = 0.53\ \text{Å}$$



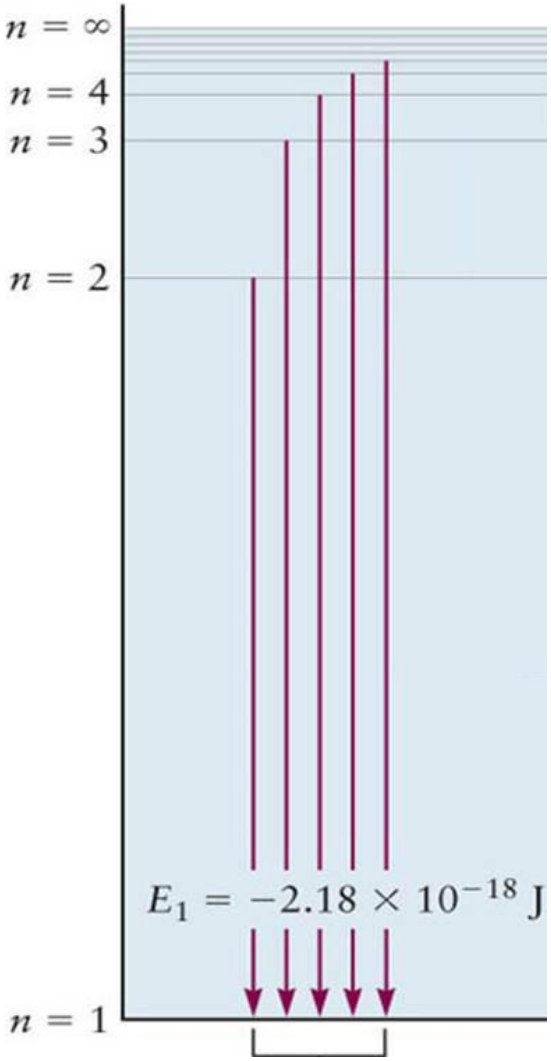


The Hydrogen Atom



Atomic Orbitals

Bohr, OGB8 p.143

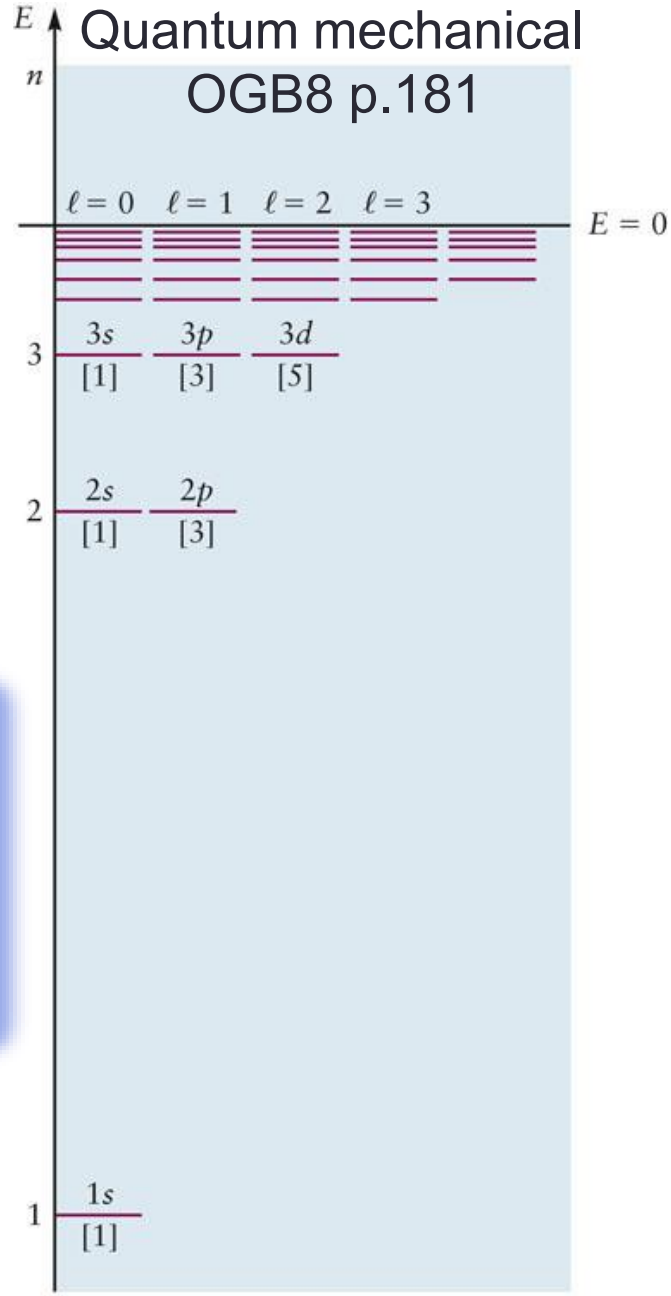


$$E_n = -\frac{\mathrm{Ry}}{n^2},$$

$$\mathrm{Ry} = 2.18\times10^{-18}\ \mathrm{J}$$

$$= 13.6\ \mathrm{eV}$$

Quantum mechanical
OGB8 p.181



Periodic Table of the Elements

1 H 1.00794																1 H 1.00794	2 He 4.002602
3 Li 6.941	4 Be 9.012182											5 B 10.811	6 C 12.0107	7 N 14.00674	8 O 15.9994	9 F 18.9984032	10 Ne 20.1797
11 Na 22.989770	12 Mg 24.3050											13 Al 26.981538	14 Si 28.0855	15 P 30.973761	16 S 32.066	17 Cl 35.4527	18 Ar 39.948
19 K 39.0983	20 Ca 40.078	21 Sc 44.955910	22 Ti 47.867	23 V 50.9415	24 Cr 51.9961	25 Mn 54.938049	26 Fe 55.845	27 Co 58.933200	28 Ni 58.6934	29 Cu 63.546	30 Zn 65.39	31 Ga 69.723	32 Ge 72.61	33 As 74.92160	34 Se 78.96	35 Br 79.904	36 Kr 83.80
37 Rb 85.4678	38 Sr 87.62	39 Y 88.90585	40 Zr 91.224	41 Nb 92.90638	42 Mo 95.94	43 Tc (98)	44 Ru 101.07	45 Rh 102.90550	46 Pd 106.42	47 Ag 107.8682	48 Cd 112.411	49 In 114.818	50 Sn 118.710	51 Sb 121.760	52 Te 127.60	53 I 126.90447	54 Xe 131.29
55 Cs 132.90545	56 Ba 137.327	57 La 138.9055	72 Hf 178.49	73 Ta 180.9479	74 W 183.84	75 Re 186.207	76 Os 190.23	77 Ir 192.217	78 Pt 195.078	79 Au 196.96655	80 Hg 200.59	81 Tl 204.3833	82 Pb 207.2	83 Bi 208.98038	84 Po (209)	85 At (210)	86 Rn (222)
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (263)	107 Bh (262)	108 Hs (265)	109 Mt (266)									

Speed of light
$c = 299\,792\,458\ \mathrm{m \cdot s^{-1}}$

Planck constant
$h = 6.626\,07 \times 10^{-34}\ \mathrm{J \cdot s}$

Vacuum permittivity
$\varepsilon_0 = 8.854\,19 \times 10^{-12}\ \mathrm{F \cdot m^{-1}}$

Elementary charge
$e = 1.602\,18 \times 10^{-19}\ \mathrm{C}$

Electron mass
$m_e = 9.109\,38 \times 10^{-31}\ \mathrm{kg}$

Gas constant
$R = 8.314\,47\ \mathrm{J \cdot mol^{-1} \cdot K^{-1}}$

Avogadro constant
$N_A = 6.022\,14 \times 10^{23}\ \mathrm{mol^{-1}}$

Boltzmann constant
$k_B = 1.380\,65 \times 10^{-23}\ \mathrm{J \cdot K^{-1}}$

Bohr radius
$a_0 = 0.529\,177 \times 10^{-10}\ \mathrm{m}$

Rydberg energy
$R_H = 13.605\,7\ \mathrm{eV}$

Standard atmosphere
$1\ \mathrm{atm} = 101\,325\ \mathrm{Pa}$

Acceleration of gravity
$g = 9.806\,65\ \mathrm{m \cdot s^{-2}}$

$$r_n = \frac{n^2}{Z} a_0$$

$$E_n = -\frac{Z^2}{n^2} R_H$$

$$L = \sqrt{l(l+1)}\,\frac{h}{2\pi}$$

Midterm 1 on Wednesday Oct. 29th

Time: 10:30 – 11:30 am